

Chelsea Knight

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Education

Department of Defense SMART Scholar

Old Dominion University | Master of Science in Data Science and Analytics

Concentration: Artificial Intelligence and Machine Learning | Expected May 2026

North Carolina State University | Bachelor of Arts in Psychology, Minor in Forensic Science

WGCA All-American Scholar | Arthur Ashe Sports Scholar Nominee | Dean's List Recipient

Technical Skills

Programming: Python (pandas, NumPy, scikit-learn, Matplotlib, Seaborn), SQL, XML, R

Databases & Big Data: MySQL, MongoDB, Oracle, Apache Spark

Machine Learning & AI: Classification models (Random Forest, SVM, Logistic Regression, Gradient Boosting), hyperparameter tuning, model evaluation (ROC-AUC, precision/recall, F1-score, cross-validation)

Data Engineering: ETL pipelines, data integration, multi-source data processing, OpenRefine

Analytics: Exploratory data analysis, statistical modeling, hypothesis testing, regression analysis

Visualization: Tableau, Observable, data storytelling

Version Control & Development: Git, GitHub, Jupyter/Google Colab

Work Experience

Office of Cost Assessment and Program Evaluation (CAPE), The Pentagon

Data Science Intern | June 2025 - August 2025

- Conducted landscape analysis of analytical tools available to analysts and explored solutions to reduce manual workflow efforts for future implementation
- Evaluated machine learning agent deployment strategies for non-technical professionals across divisions, supporting early-stage AI integration planning
- Gained hands-on experience with Oracle databases and Tableau integration for defense analytics applications
- Facilitated cross-divisional communication by documenting analytical workflows, tool usage, and identifying functional solutions to improve operational efficiency

Prince George's County Department of Social Services

Quality Assurance & Compliance Coordinator | Feb 2022 - July 2025

Community Developer I—Data Analyst | Dec 2019 - Feb 2022

Quality Assurance Data Specialist | Apr 2019 - Dec 2019

- Managed agency fiscal budget analysis by collecting data from multiple databases, calculating projections, and presenting trend analysis with actionable insights to Office of Management and Budget (OMB) for County Council stakeholders
- Conducted quantitative and qualitative data analysis to support agency compliance with COMAR policies and federal regulations, generating reports for county and state budget allocation
- Designed and maintained data tracking systems using SQL, JavaScript, and spreadsheet automation for CPS clearance processing and case management
- Collaborated with Maryland DSS work groups to modify review processes for Child, Juvenile and Adult Management Systems (CJAMS) implementation
- Performed data extraction, transformation, and validation across multiple database applications to ensure data integrity for presentations and compliance audits

Research & Projects

Predicting Employee Attrition Using Machine Learning | Python, Scikit-learn, Pandas

- Built end-to-end ML pipeline comparing Logistic Regression, Decision Trees, Random Forest, SVM, and Gradient Boosting on IBM HR dataset; achieved ROC-AUC of 0.815 with tuned SVM through GridSearchCV optimization and identified compensation and tenure as key turnover predictors through feature importance analysis

Analyzing Housing Insecurity in the United States | Python, Observable, Data Visualization

- Conducted exploratory analysis of U.S. housing insecurity trends (2016-2020) using multiple datasets, building interactive choropleth maps, multi-line charts, and scatterplots in Observable to communicate insights on shelter availability and population trends across states

Movie Recommendation System | PySpark, ALS, Python

- Developed hybrid recommendation engine combining ALS collaborative filtering (RMSE 0.89), content-based filtering with cosine similarity, and Bayesian average popularity ranking to generate personalized movie suggestions from MovieLens 100K dataset using distributed processing with Apache Spark

Mortality and Disability Predictive Modeling | Python, Pandas, Scikit-learn, Data Integration

- Integrated three CDC datasets (disability prevalence, mortality by age, cause of death) through ETL processes including data mapping and schema alignment; developed Random Forest model ($R^2 = 0.93$) predicting mortality patterns by age group, disability type, and region in collaborative team environment

NC State Undergraduate Research Assistant | 2017 - 2018

Conducted research on criminal justice data including recidivism trend analysis for NIDA parolee intervention methods, coded qualitative police training interviews, and performed regression analysis on Wake County Jail mental health indicators to examine sexual violence recidivism patterns.